



Friedrich-Alexander-Universität  
Erlangen-Nürnberg

# Compton Effect

[ALC-B22]

Department of Physics

Lab Report

M766

Name: Saujanya Dubale, Kavan Purohit

Matriculation Number: 23148509, 23162718

Supervisor: Benedetta Bruno

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# 1 Introduction

The Compton effect, discovered by Arthur Holly Compton in 1921, is a fundamental phenomenon in quantum mechanics that demonstrates the particle nature of light. This groundbreaking discovery earned Compton the Nobel Prize in Physics in 1927.

Compton's experiments involved scattering X-rays on various targets, including graphite and molybdenum. He observed an unexpected result: the scattered X-rays had a longer wavelength than the incident X-rays. This wavelength shift could not be explained by classical electromagnetic theory, which treated light purely as a wave.

## Compton's Original Experimental Setup

The significance of the Compton effect extends beyond its initial discovery:

1. Particle Nature of Light : It provided crucial evidence for Einstein's photon theory of light, supporting the concept that light can behave as both a wave and a particle.
2. Quantum Mechanics Validation : The effect demonstrated the quantized nature of light-matter interactions, a cornerstone of quantum mechanics.
3. Conservation Laws : It showed that energy and momentum are conserved in individual photon-electron collisions, not just on average as proposed by some competing theories.
4. Technological Applications : Understanding the Compton effect has led to advancements in medical imaging (e.g., Compton scattering tomography) and radiation shielding techniques.

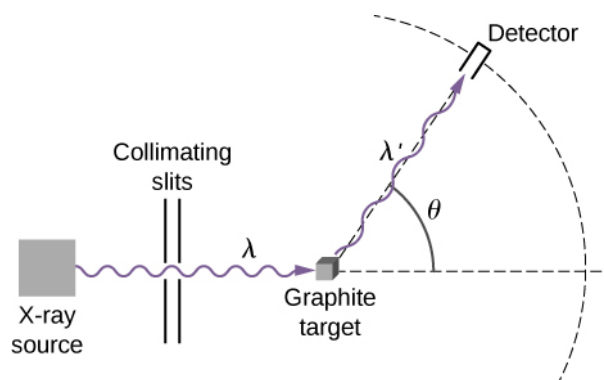


Figure 1: Experimental setup for studying Compton scattering.

## Historical Context and Competing Theories

In the early 20th century, the nature of light was a subject of intense debate. While Maxwell's equations had successfully described light as an electromagnetic wave, phenomena like the photoelectric effect (explained by Einstein in 1905) suggested a particle-like nature.

Compton's discovery was met with both excitement and skepticism. A competing classical explanation was proposed by Niels Bohr, Hendrik Kramers, and John Slater in 1924. Their theory, known as the BKS theory, attempted to explain the Compton effect using virtual oscillators and a statistical interpretation of energy conservation.

However, the BKS theory was soon disproven by experiments conducted by Walther Bothe and Hans Geiger in 1925. Using coincidence counting techniques, they showed that energy and momentum were conserved in individual scattering events, not just statistically.

## 2 Theoretical Background

### 2.1 Derivation of the Compton Scattering Formula

The Compton effect explains the interaction between a photon, with momentum  $\vec{p}_\gamma$ , and a quasi-free electron. During this interaction, the photon collides with the electron and transfers some of its energy, giving the electron a new momentum  $\vec{p}_e'$ . Consequently, the photon's wavelength increases, which means it loses energy, and this shift in wavelength depends on the scattering angle  $\phi$ . This phenomenon demonstrates the particle-like properties of photons, showing that light can transfer momentum to electrons, thereby supporting the quantum nature of light.

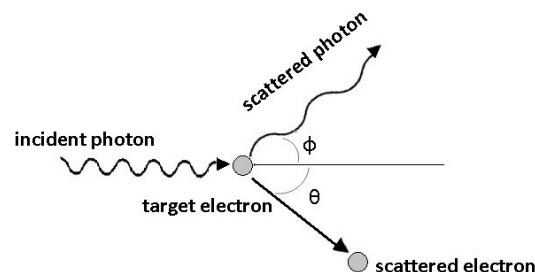


Figure 2: Compton Scattering

Let's go through the derivation in more detail:

1. Initial State:

Photon: Energy

$$E^1 = h\nu_1$$

Momentum

$$p^1 = \frac{h}{\lambda^1}$$

Electron: At rest, Energy

$$E_0 = mc^2$$

2. Final State (after scattering):

Photon: Energy

$$E_2 = h\nu_2$$

Momentum

$$p_2 = h/\lambda_2$$

Electron: Energy

$$E = \sqrt{((pc)^2 + (mc^2)^2)}$$

3. Conservation of Energy:

$$E_1 + E_0 = E_2 + E$$

$$h\nu_1 + mc^2 = h\nu_2 + \sqrt{((pc)^2 + (mc^2)^2)}$$

4. Conservation of Momentum (x and y components):

$$x: h/\lambda_1 = (h/\lambda_2)\cos\theta + p\cos\phi$$

$$y: 0 = (h/\lambda_2)\sin\theta - p\sin\phi$$

6. Final Result:

After squaring and combining the momentum equations, and using the energy conservation equation, we can eliminate p and  $\phi$ .

$$\lambda_2 - \lambda_1 = (h/mc)(1 - \cos\theta)$$

This can be rewritten as:

$$\Delta\lambda = \lambda c(1 - \cos\theta)$$

Where  $\lambda c = h/mc$  is the Compton wavelength of the electron.

what happens at  
theta=180 and  
theta=0???

## 2.2 Cross Sections

Cross sections are crucial measures in scattering experiments, providing information about the probability of interaction between particles.

### 1. Differential Cross Section ( $d\sigma/d\Omega$ ):

- Describes how the intensity of scattered radiation varies over a solid angle  $\Omega$
- Formula:

$$\frac{d\sigma}{d\Omega} = \left(\frac{1}{\xi}\right) \cdot \left(\frac{R}{I}\right) \cdot \left(\frac{1}{\rho d}\right) \cdot \left(\frac{A_u}{Z}\right) \cdot \left(\frac{1}{d\Omega}\right)$$

Where:

- $\xi$  is the detector efficiency
- $R$  is the counting rate
- $I$  is the incident intensity
- $\rho$  is the target density
- $d$  is the target thickness
- $A_u$  is the atomic mass unit
- $Z$  is the atomic number
- $d\Omega$  is the solid angle element

### 2. Total Cross Section ( $\sigma$ ):

- Obtained by integrating the differential cross-section over the entire solid angle
- Formula:

$$\sigma = \int \left(\frac{d\sigma}{d\Omega}\right) d\Omega$$

The cross-section has units of area and is often expressed in barns (1 barn =  $10^{-28}$  m<sup>2</sup>).

## 2.3 Interaction of Gamma Rays with Matter

Gamma rays interact with matter through three primary processes:

### 1. Photoelectric Effect:

- Dominant at low energies (typically below 100 keV for most materials)
- The entire photon energy is absorbed by an atomic electron, which is then ejected
- Cross-section:

$$\sigma_{Photo} \propto \frac{Z^5}{E_\gamma^{3.5}}$$

- Strongly dependent on atomic number, making high-Z materials effective for shielding

### 2. Compton Scattering:

- Dominant at intermediate energies (typically 100 keV to 10 MeV)
- Photon scatters off an electron, losing some energy and changing direction
- Cross-section:

$$\sigma_{Compton} \propto \frac{Z}{E_\gamma} \cdot \ln(E_\gamma)$$

- Less dependent on Z compared to photoelectric effect

### 3. Pair Production:

- Occurs at energies above 1.022 MeV (twice the electron rest mass)
- Photon converts into an electron-positron pair in the presence of a nucleus
- Cross-section:

$$\sigma_{Pair} \propto Z^2 \cdot \ln(E_\gamma)$$

- Becomes dominant at high energies

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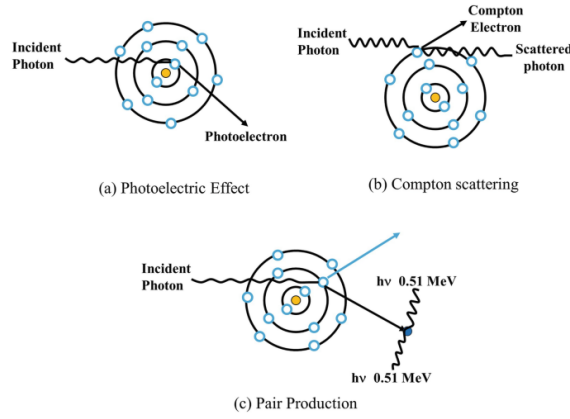


Fig 3: Representation of  $\gamma$  interactions with matter

The attenuation of photons in matter follows the exponential law:

$$N(x) = N(0) \exp(-\mu x)$$

Where:

- $N(x)$  is the number of photons at depth  $x$
- $N(0)$  is the initial number of photons
- $\mu$  is the linear attenuation coefficient

The linear attenuation coefficient  $\mu$  is the sum of the coefficients for each interaction process:

$$\mu = \mu_{photo} + \mu_{compton} + \mu_{pair}$$

## 2.4 Klein-Nishina Formula

The Klein-Nishina formula, derived by Oskar Klein and Yoshio Nishina in 1929, provides a quantum mechanical description of the differential cross-section for Compton scattering:

$$\left(\frac{d\sigma}{d\Omega}\right) = \left(\frac{r_0^2}{2}\right) \cdot \left(\frac{\lambda'}{\lambda}\right)^2 \cdot \left(\frac{\lambda}{\lambda'} + \frac{\lambda'}{\lambda} - \sin^2\theta\right)$$

here:

- $r_0$  is the classical electron radius
- $\lambda$  and  $\lambda'$  are the wavelengths of the incident and scattered photons

-  $\theta$  is the scattering angle

This formula accounts for relativistic effects and reduces to the classical Thomson scattering formula at low energies.

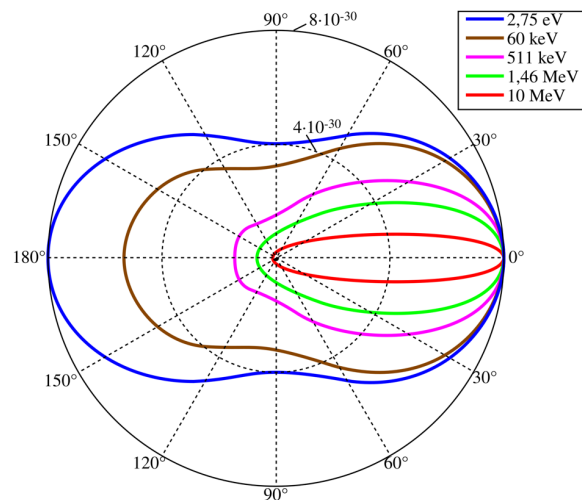


Figure 6: Klein–Nishina distribution of scattering-angle cross sections over a range of commonly encountered energies

## 3. Experimental Setup and Methodology

### 3.1 Scintillation Detector

Scintillation detectors are widely used for gamma-ray spectroscopy due to their high efficiency and good energy resolution. The main components are:

#### 1. Scintillator Material:

- Typically  $NaI(Tl)$  or  $CsI(Tl)$  for gamma-ray detection
- Emits light when excited by ionizing radiation
- Light output is proportional to deposited energy

#### 2. Light Guide:

- Transmits scintillation light to the photomultiplier tube
- Designed to minimize light loss

### 3. Photomultiplier Tube (PMT):

- Converts light into an electrical signal
- Consists of a photocathode, dynodes, and an anode
- Provides signal amplification through electron multiplication

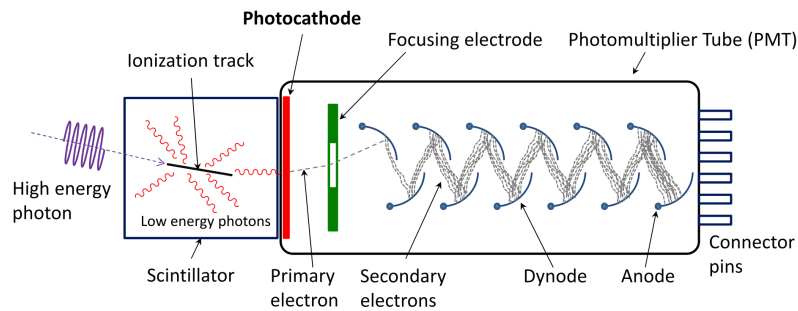


Figure 7: Schematic of a typical Scintillation Detector

The process of detection involves:

1. Gamma-ray interaction in the scintillator
2. Light emission from the scintillator
3. Light transmission to the PMT
4. Photoelectric conversion at the photocathode
5. Electron multiplication in the dynode chain
6. Signal collection at the anode

Why does one get a linear function between the anode signal of the photo tube and the impinging photon energy?

Explain the terms photo-peak, Compton-continuum and Compton-edge and sketch the spectrum of a monochromatic  $\gamma$ -line.

## 3.2 Signal Processing

The experimental setup includes various electronic modules for signal processing:

### 1. Preamplifier:

- Provides initial amplification close to the detector
- Improves signal-to-noise ratio

### 2. Amplifier:

- Further amplifies and shapes the signal

- Typically provides both linear and logarithmic outputs

### 3. Discriminator:

- Generates a logic pulse when the input signal exceeds a threshold
- Used for timing and triggering

### ~~4. Single Channel Analyzer (SCA):~~

- ~~- Selects pulses within a specific energy range~~
- ~~- Outputs a logic pulse for qualifying events~~

### ~~5. Multichannel Analyzer (MCA):~~

- ~~- Digitizes and sorts pulses by amplitude~~
- ~~- Produces energy spectrum histograms~~

Linear gate stretcher  
and ADC explanation  
are missing

### 6. Coincidence Unit:

- Detects simultaneous events from multiple detectors
- Essential for background reduction and angular correlation studies

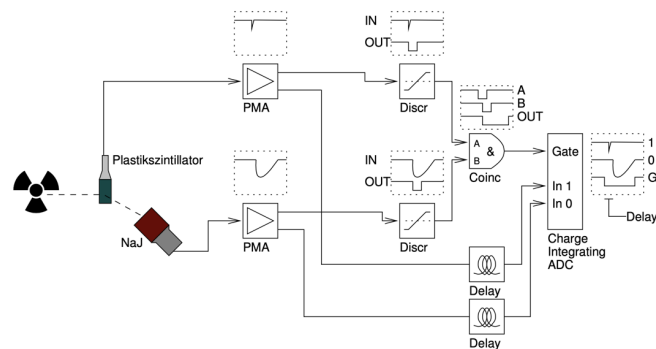


Figure 8: Experimental Setup

### ~~3.3 Data Acquisition and Analysis~~

~~The experiment will involve:~~

#### ~~1. Energy Calibration:~~

- ~~Using known gamma-ray sources (e.g.,  $^{137}\text{Cs}$ ,  $^{60}\text{Co}$ )~~
- ~~Establishing relationship between channel number and energy~~

#### ~~2. Background Measurement:~~

- ~~Recording spectrum without source to account for environmental radiation~~

#### ~~3. Compton Scattering Measurements:~~

- ~~Varying scattering angle~~
- ~~Recording energy spectra at each angle~~

#### ~~4. Data Analysis:~~

- ~~Peak fitting to determine scattered photon energies~~
- ~~Calculation of wavelength shift~~
- ~~Determination of Compton wavelength~~
- ~~Cross-section calculations~~

#### ~~5. Error Analysis:~~

- ~~Statistical uncertainties in peak positions~~
- ~~Systematic errors in angle measurements and detector efficiencies~~

#### ~~6. Comparison with Theory:~~

- ~~Plotting experimental results against Klein Nishina predictions~~
- ~~Discussion of any discrepancies~~

## ~~4. Expected Results and Discussion~~

~~We anticipate observing:~~

- ~~1. A clear shift in the energy of scattered photons consistent with the Compton formula~~
- ~~2. A distribution of scattering angles following the Klein Nishina prediction~~
- ~~3. Measured cross sections that align with theoretical calculations~~

~~Potential sources of error and their mitigation strategies include:~~

- ~~Energy Resolution: Use of high quality detectors and careful calibration~~
- ~~Background Radiation: Proper shielding and background subtraction~~
- ~~Geometric Effects: Precise alignment of source, target, and detector~~
- ~~Multiple Scattering: Use of thin targets to minimize this effect~~

~~The experiment will provide valuable hands on experience with:~~

- ~~Advanced radiation detection techniques~~
- ~~Signal processing and data acquisition systems~~
- ~~Data analysis and error propagation methods~~
- ~~Comparison of experimental results with theoretical models~~

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